Web scraping case study

Industrial equipment catalog: 6,153 products from 56 categories into clean CSV files

6,153	56	100%	~35 min
products collected	categories	of category product counts matched the site	full unattended run

The task

A client needed the product catalog of a large Russian-language industrial pump website (about 60 categories, roughly 8,000 products) turned into spreadsheet files. Each product had to include its own description and technical specs. A listing page alone was not enough, because the unique description and the labelled specs live on each product card. The client also asked for the scraper itself and a short how-to, so they could re-run it later.

What I built

- **Python scraper** (requests + BeautifulSoup) that reads each category listing, opens every product card and extracts name, model, source URL, category, full description and six specs: flow, head, power, RPM, weight and dimensions.
- **Parsing by label, not by position.** Specs are matched by their on-page labels, so products with missing fields, extra fields or a different field order still parse correctly. The source site misspells one label, and the scraper handles that too.
- **Resumable runs.** One CSV is written per category as soon as it finishes. If the run is interrupted, re-running the same command skips finished categories and continues.
- **Fault isolation and polite crawling.** A failing page is logged and skipped without stopping the run; requests are spaced out to avoid loading the target site.
- **Excel-friendly CSV.** Semicolon-delimited, UTF-8 with BOM, so Cyrillic text and multi-line descriptions open correctly in Excel and import cleanly into a CMS. A small merge script combines category files into one.

How I verified the data

Before the full run I counted the products in every category from the listing pages. After the run I compared those counts with the rows in each output file: all 56 categories matched exactly. I then audited every file for empty names, empty descriptions, products without specs and duplicate links, and checked sample product pages in five different categories to confirm the page layout was consistent.

Edge cases found and reported to the client

- 24 products have no description and 15 have no specs on the source site itself. I spot-checked this on the live pages and left those cells empty instead of guessing.
- One category (47 sucker-rod pumps) publishes no structured specs at all, only a text description.
- The first delivery was a 93-product test category so the client could approve the format before the full run.

Sample output

Model	Flow, m³/h	Head, m	Power, kW	RPM	Weight, kg	Dimensions, mm
12НДСР	1080	68	250	1450	1174	1358×1392×1020
2ГНОМ 10-10	10	10	1.1	2900	20	210×210×420
ЦНП-100/40	100	40	30	1450	382	800×600×1100

Rows are real records from the project; units are translated here for readability. The delivered files keep the source text, and the full descriptions are in a separate column.

Tools

Python 3, requests, BeautifulSoup4, csv. Client identity omitted; a scraper script and a data sample are available alongside this case study.